

Supplementary Information for

Frequency-Enhanced Wavelet-based Transformer in Imitation Learning for Humanoid Robot

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Project Website: <https://humanoid-black-knight.github.io/>.

Supplementary material

Fig S1: Visualization of the dynamic weight α in the FE-EMA module (training phase, effect of 2000 epoch under different simulation tasks).

Fig S2: The design of humanoid robot (*Humanoid Black Knight*, HBK).

Table S1: Comparison of Parameters and FLOPs for One Epoch.

Table S2: Hyperparameters of the FEWT (based on ACT Policy).

Video S1: A collection of experiments on humanoid robots

A. Comparison of Parameters and FLOPs

Table S1 shows that, based on the ACT Policy baseline and Backbone (ResNet18), the FE-EMA module reduces the computational complexity (FLOPs) compared to the EMA module. Where the Simulation Tasks (ACT baseline) are *Cube Transfer* and *Bimanual Insertion*, respectively.

Table S1. Comparison of Parameters and FLOPs for One Epoch.

Method	Backbone	Simulation Tasks	
		#.Param.	FLOPs
Baseline (ACT)	ResNet18	60.7566 M	37570.51 M
EMA + ACT		60.7592 M	37616.59 M
FE-EMA + ACT (ours)		60.7621 M	37596.91 M

B. Training Details

We adopt the hyperparameters reported in the ACT Policy paper, with the sole modification of reducing the batch size from 8 to 4. The complete hyperparameter settings are summarized in **Table S2**. We deploy our policy with inference on a desktop with an NVIDIA RTX 3060 GPU.

Table S2. Hyperparameters of the FEWT (based on ACT Policy).

Hyperparameter	FEWT
Learning rate	1e-5
Batch size	4
# encoder layers	4
# decoder layers	7
Feedforward dimension	3200
Hidden dimension	512
# heads	8
Chunk size	100
KL weights	10

For the two simulation tasks, *Cube Transfer* and *Bimanual Insertion*, the dynamic weight α is consistently maintained at approximately 0.5 during training (see **Fig. S1**). This suggests that the cross-spatial and frequency-domain feature weights are well balanced during model training. Dynamically fusing the time-domain and frequency-domain features allows effective capture of information across different scales.

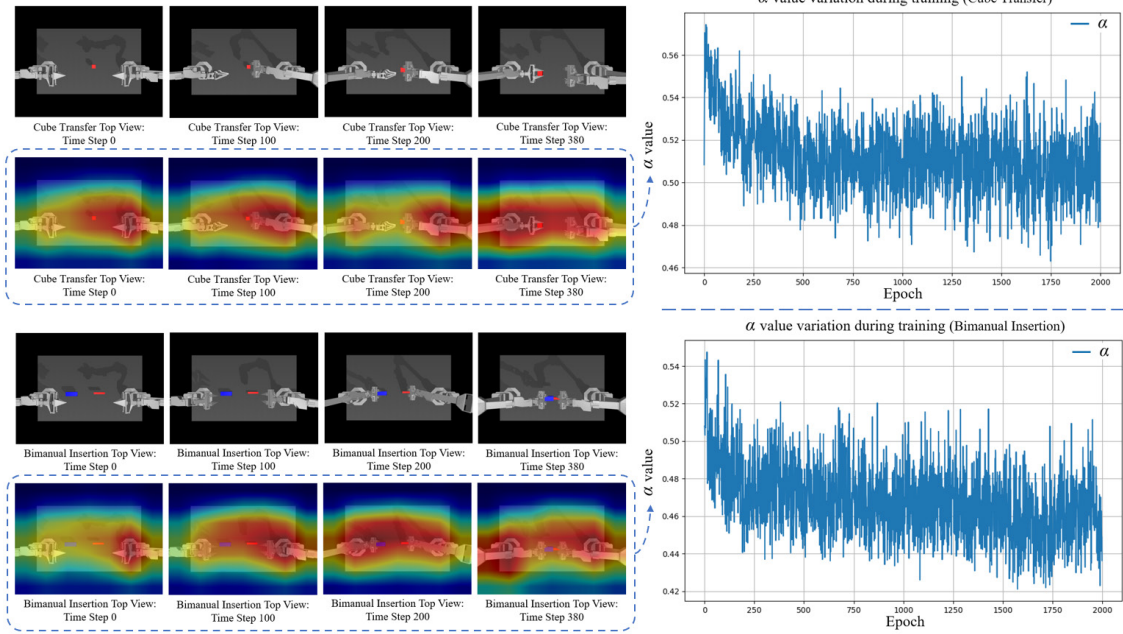


Fig. S1: Visualization of the dynamic weight α in the FE-EMA module (training phase, effect of 2000 epoch under different simulation tasks).

C. The Construction of Humanoid Black Knight

The construction of *Humanoid Black Knight* (HBK) mainly consists of two 7-dof (arm + gripper) robotic arms (*ViperX-300*), a two-wheeled differential mobility chassis (*Diablo*), three RGB cameras (Logitech C922x webcams), and an inertial measurement unit (IMU sensor). The purpose of the chassis linkage restraint parts is to prevent skidding during chassis steering movements (See **Fig. S2**). The video collection demonstrating the humanoid robot can be found in **Video S1**.

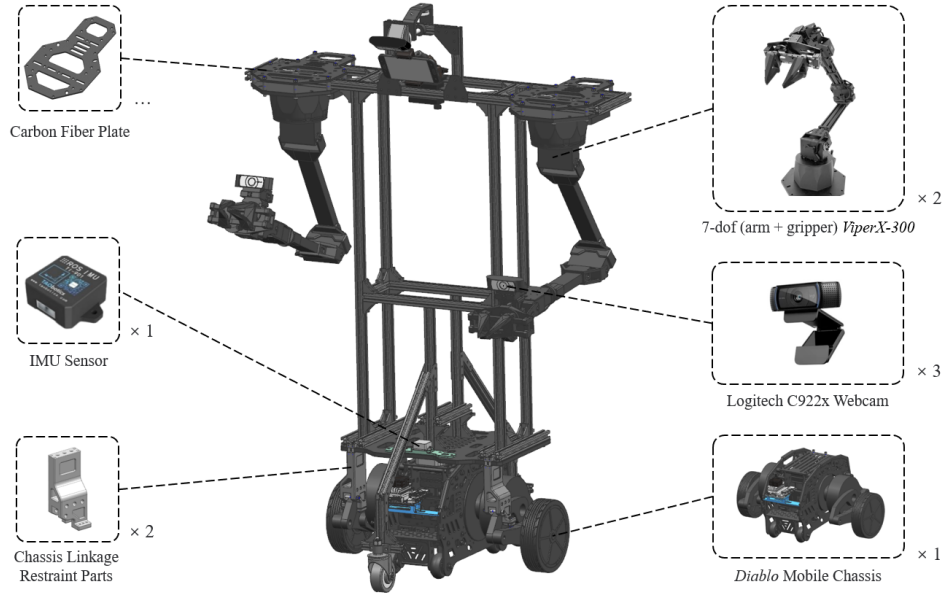


Fig. S2: The design of humanoid robot (*Humanoid Black Knight*, HBK).